Amber Handal

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EDUCATION

Northwestern University — Evanston, IL Sep 2025 – Dec 2026
M.S. in Robotics

University of Florida — Gainesville, FL May 2019 – May 2024
B.S. in Computer and Information Science Engineering

SKILLS

Software	Python, C, C++, Embedded C, TypeScript, SQL, Git, CI/CD, AWS, Docker
Robotics	ROS/ROS 2, SLAM, Motion Planning, Kinematics, State Estimation
Perception & ML	Computer Vision, Deep Learning, Reinforcement Learning, On-Device Inference
Embedded Systems	Microcontroller Architecture, Peripherals (PWM, ADC, I2C, SPI, UART), PID Control, Brushed DC Motors, Sensor/Actuator Interfacing, Real-Time Systems, PCB Design
Systems & Low-Level	Linux Kernel Concepts, Drivers, Interrupts, Concurrency, Memory Layout, GCC Toolchain, Debugging (JTAG, Logic Analyzer, QEMU), Custom Linking

WORK EXPERIENCE

Software Engineer Infotech — Gainesville, FL Sep 2022 – Aug 2025

- Built automated PDF pipelines using Python, TypeScript, and AWS Lambda to retrieve and classify plan data.
- Mentored three engineers to extend the system for DOT clients using token-based authentication and S3 APIs.
- Conducted code reviews and optimized Lambda concurrency and database queries for improved scalability.

PROJECTS

WatchDog — Autonomous Quadruped Inspection Jan 2026 – Present

- Building autonomous inspection system on Unitree Go2 with Nav2 and custom velocity command bridge.
- Configuring RTAB-Map visual + ICP registration for SLAM using Unitree L2 LiDAR and RealSense D435i.
- Resolving multi-machine ROS 2 issues including DDS discovery, TF timestamp sync, and QoS tuning.
- Implementing frontier-based exploration for autonomous mapping with 2D LaserScan obstacle avoidance.
- Integrating SAM 3 on external GPU, projecting segmentation masks to 3D for object change detection.

PenPal — Vision-Guided Conversational Writing Robot Dec 2025

- Co-developed a ROS 2 system for a 7-DoF Franka arm to read and write on a moving whiteboard with 3 peers.
- Led perception using RealSense RGB-D, AprilTags, and OpenCV to estimate board pose and publish TF.
- Integrated a Gemini vision-language model for OCR + response generation via a ROS 2 service interface.
- Supported Cartesian writing execution by aligning the pen TCP to the board frame for consistent strokes.
- Generated font-based toolpaths by parsing OTF files to produce multi-language writing trajectories in 3D space.

Aggro-Bots — Biometric-Responsive Swarm Robotics Game Nov 2025 – Dec 2025

- Developed an embedded C control stack on Micro:bit (nRF52) with joystick input and heartbeat telemetry.
- Implemented a lightweight wireless protocol with serialized messages and role-based swarm coordination.
- Wrote low-level drivers for SAADC, GPIO, and I^2C motor control to drive chassis at deterministic rates.
- Built closed-loop pursuit and avoidance behaviors using distributed heuristics and filtered velocity commands.

Critter Collector — Educational Mobile Game Aug 2023 – May 2024

- Deployed backend in Node.js and MongoDB for an educational location-based game inspired by Pokémon Go.
- Migrated code from Unity to Unreal Engine 5 using JavaScript and C++ to enhance performance and scalability.
- Partnered with a 5-member cross-functional team, ensuring consistent communication and sprint alignment.